Shade Rahman

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering, GPA: 4.0 — *Aug. 2024 – May 2028*

Experience

Solutions Engineer Intern — Aug. 2026 – Present
PeakActivity — *Orlando, FL / Remote*

- Automated pharmaceutical intake end to end by building a **35-block** SteelEngine workflow (document extraction, confidence-based routing, human review), validated across **6 acceptance scenarios**.
- Addressed campaign peaks skewing alerts by prototyping a SteelEngine **anomaly-reporting tool** with hybrid baselines and materiality gates, calibrated on **8 dates** (**0 to 3** incidents on quiet days, **8** on a high-signal day).
- Identified the fastest path to an executable automation by benchmarking **3 workflow-authoring approaches** across **9 People Ops workflows** (**218 blocks**, **234 edges**); structured raw-transcript input won.
- Automated test-case drafting from Jira tickets with a proof-of-concept **deterministic compiler** that outputs versioned Playwright cases, restricted to allowlisted HTTPS actions and rejecting checkout and payment steps.

President — May 2026 – Present
Bengali Student Association — *Orlando, FL*

- Enabled **15 planned events** to run in parallel without adding headcount by splitting a **9-person board** into dedicated committees (Outreach, Dance) with delegated ownership.
- Contributed to a **110%** year-over-year increase in average per-tabling revenue by building an **inventory, sales log, and profit tracker** that let the Outreach Committee run weekly tablings independently.
- Secured **$1,883.88** in itemized funding approval for the first event, out of a **$3,700** SGA budget across **3 flagship events**, through vendor negotiation, compliance documentation, and multi-form procurement requests.

UI/UX Designer — Apr. 2026 – Present
Knight Hacks — *Orlando, FL*

- Contributed to a **41%** increase in applications (**1,000+** expected applicants) by designing **wireframes, landing page layouts,** and **conversion-focused UI flows** for Knight Hacks IX.
- Kept developer handoff clean across a **17-person** design team by maintaining a shared **Figma** and **Notion** design system.

Projects

FinePrint (SASEhack 2026 Winner) | *Next.js, TypeScript, FastAPI, PyMuPDF, OpenAI API, Vercel* — Sept. 2026

- Won **Best Finance Hack** with a financial-aid letter analyzer that isolated **$16,900** of gift aid from **$45,400** advertised on a sample letter, verifying every quote and amount against the PDF with a deterministic **evidence gate**.
- Made scenarios recompute instantly with no API call by keeping all financial math in **pure TypeScript** in the browser, backed by **87 API tests**, **53 web tests**, and a **5-letter** replay corpus with **0 differences** from its answer key.
- Held live extraction cost to about **$0.06** per synthetic stress letter by routing to a balanced-cost model by default and a stronger fallback only on validation failure, with SDK retries disabled.

Rescuebot (ShellHacks 2026) | *Python, FastAPI, asyncio, ROS 2, Docker, ESP32, Raspberry Pi 5* — Sept. 2026

- Built the **web dashboard and driving software** for a 4-person search-and-rescue robot that spots people with an AI camera, **demoed live to judges**, with browser driving and a LiDAR-built map.
- Made Pi-to-ESP32 control reliable with a **JSON serial protocol** (sequence numbers, stale commands dropped) checked by **35 shared test vectors**, and fixed I2C stalls from **8.6 ms** transactions (**38/38 ACKs**, **53 ms** longest silence).
- Integrated **ROS 2 Jazzy** (SLAM Toolbox, Nav2) in **Docker** with autonomy routed through the dashboard's safety checks, and fixed wheel stalls under Nav2 with a minimum PWM and one-axis-at-a-time driving.

Bolo | *Next.js, TypeScript, Tesseract.js, Gemini API, OpenAI API, Vercel* — June 2026

- Cut drafting time for a real business email thread from **30 to 4 minutes** (self-timed) by building a **Next.js** app that turns native-language text, speech, and email screenshots into English drafts.
- Preserved embedded English brand names, phrases, and numbers through Bangla, Spanish, and Gujarati translation, verified by **35 regression tests**.
- Hardened the app with prompt-injection detection and Unicode-aware OCR cleanup, backed by **97 offline tests**, and fixed **10 audit bugs**, including a critical compose-mode refine failure.

Technical Skills

Languages: C, C++, Python, Java, TypeScript, JavaScript, ARM Assembly
Frameworks & APIs: Next.js, React, FastAPI, Flask, Flutter, Tailwind CSS, REST APIs, Gemini API, OpenAI API
Tools & Platforms: Git, Docker, Vercel, Supabase, MATLAB, Figma, Notion, Adobe Photoshop, Canva, Microsoft Office
Hardware & Systems: Raspberry Pi, ESP32, ROS 2, I2C, FPGA, Vivado, RTL Design, Autodesk Inventor, Autodesk Revit
Awards: President's Honor Roll, Pegasus Gold Scholar, Bright Futures Florida Academic Scholar, Eda & Cliff Viner Scholar.